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Thesis for Master's Degree

**Acoustic imaging with three-dimensional acoustic
Luneburg metalens**

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School of Mechanical Engineering

Gwangju Institute of Science and Technology

2022

Acoustic imaging with three-dimensional acoustic Luneburg metalens

3차원 음향 룬버그 메타렌즈를 사용한
음향 이미징

Acoustic imaging with three-dimensional acoustic Luneburg metalens

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A thesis submitted to the faculty of the Gwangju Institute of Science and Technology in partial fulfillment of the requirements for the degree of Master of Science in the School of Mechanical Engineering.

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ABSTRACT

Luneburg lens is a type of aberration-free gradient index lens that focuses or collimates waves at all incident direction. Due to its characteristics, the lens has been mainly used in communication fields such as high-gain antennas. Metamaterials refer to materials with unique properties that do not exist in nature and are designed by periodically arranging scatterers (or meta-atoms) smaller than the wavelength. Therefore, the Luneburg metalens is realized by periodically arranging meta-atoms so that the gradient refractive index of the lens is achieved. Both concepts were first proposed in the optic/electromagnetic realm but were extended to mechanical waves due to their similarity to the elastic/acoustic realm. In this paper, three-dimensional acoustic Luneburg metalens worked in water is introduced. Since the properties and performance of the lens can be identified through the effective material properties, these properties were calculated through band structure calculations and scattering parameter retrieval method for meta-atoms. The full width at half maximum, indicating the focusing performance, was calculated, and the numerical and experimental results for various incident directions were compared to confirm the omnidirectional characteristics. As a result, the newly proposed devices have the potential to be used for next-generation sonar, non-destructive testing, or medical ultrasonic sensors in underwater communication and imaging fields.

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ABSTRACT (Korean)

룬버그 렌즈는 수차가 없는 구배 굴절률 렌즈의 한 종류로서 모든 입사 방향에 대해 파동을 집중시키거나 시준할 수 있다. 그 특성으로 인해 고이득 안테나와 같은 통신 분야에서 주로 활용되었다. 메타물질은 자연계에 존재하지 않는 특이한 특성을 가진 물질을 뜻하며, 파장보다 작은 산란체들 (또는 메타원자들)을 주기적으로 배치함으로써 설계된다. 따라서 룬버그 메타렌즈는 렌즈의 구배 굴절률이 달성되도록 메타 원자를 주기적으로 배열함으로써 구현된다. 두 개념 모두 광/전자기 영역에서 먼저 제안되었지만, 탄성/음향 분야와의 유사성으로 역학파로 확장되었다. 본 논문에선 수중에서 작동되는 3차원 음향 룬버그 메타렌즈가 소개된다. 유효 물성치를 통해 그 렌즈의 특성 및 성능을 파악할 수 있기 때문에 메타 원자에 대하여 밴드 구조 계산과 산란 매개변수 검색 방법을 통해 물성치들이 계산됐다. 집중 성능을 나타내는 반치전폭이 계산되었고, 전 방향 특성을 확인하기 위해 여러 입사 방향에 대한 수치 및 실험 결과가 비교되었다. 결과적으로 새롭게 제안된 장치는 수중 통신 및 이미징 분야의 차세대 소나, 비파괴 검사 또는 의료용 초음파 센서에 활용될 잠재성을 지닌다.

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TABLE OF CONTENTS

ABSTRACT	i
ABSTRACT (Korean)	ii
TABLE OF CONTENTS	iii
Chapter 1. INTRODUCTION	1
Chapter 2. UNDERWATER ULTRASOUND IMAGING.....	2
2.1 Refractive indices and transformation optics	2
2.2 Dispersion relation and designed metalenses.....	4
2.3 Retrieved values	7
2.4 Underwater ultrasound imaging.....	9
Chapter 3. SUPPLEMENTARY INFORMATION.....	12
3.1 Numerical simulations	12
3.2 Fabrication.....	15
3.3 Experimental setup, method, and measured signals	16
Chapter 4. DISCUSSION	17
REFERENCE	18
ACKNOWLEDGEMENT.....	24

Chapter 1. INTRODUCTION

An aberration-free spherical Luneburg lens [1] is a gradient index (GRIN) lens that concentrates propagating waves to the opposite surface of the lens. The lens was modified to generate an inner focus and an outer focus [2], and it was numerically generalized [3]. The lens is not perfect [4] because of wavelength limitations [5]; however, it is advantageous over conventional lenses because the focus does not suffer from aberrations at all incident angles. Research has been conducted for applications such as radar detection and antennas that require wide-angle scanning and high-gain radiation performance [6–12] due to their excellent focusing, beam steering, and collimation capabilities.

With the advent of metamaterials, extensions into the acoustic realm, and applications of transformation optics (TO) [13–22], lenses have been becoming more versatile in structure and their focusing capabilities have been much improved since they were first proposed. Further, research on GRIN devices with phononic crystals (PC) has inspired interest in focusing lenses [23–29], and two- (2D) and three-dimensional (3D) acoustic Luneburg metamaterial lenses have been developed for the air and water environment [30–38]. Structural GRIN lenses were also studied [39–41]. Unlike superlenses [4], which can achieve subwavelength imaging due to their negative refraction, these Luneburg metamaterial lenses have positive material properties. Their inclusion structures, which determine the degree of anisotropy and effective parameters, are suitable for broadband applications because of their nonresonant properties [27]. The main advantage of these acoustic metamaterial lenses over optical and electromagnetic metamaterial lenses is that they can be used in water [26,28,32,33].

Ultrasonic transducers have advantages of low radiation exposure, portability, low cost compared to other imaging systems, and real-time inspection [43]. However, they require a phased array system, are challenging for detecting deep objects, and there is a risk of misdiagnosis because the ultrasound imaging performance varies based on the direction of the transducer. Further, it cannot detect objects smaller than the wavelength of ultrasonic waves, limiting the resolution. It is rare for GRIN

metamaterial lenses to be implemented in the underwater [32,33] or ultrasonic domain [42] compared to the acoustic domain in air. Thus, the underwater acoustic lenses that can obtain subwavelength ultrasound imaging in a wide field of view (FOV) should be studied to address these problems [35,43–45].

Chapter 2. UNDERWATER ULTRASOUND IMAGING

2.1 Refractive indices and transformation optics

An acoustic spherical Luneburg lens focuses acoustic waves or transforms point sources into plane waves without aberration in all directions. However, it is not easy to apply this lens directly to devices because of their large volume. Therefore, a spherical Luneburg lens, which is attributed to $n(r) = \sqrt{2 - (r/R)^2}$ (where $n(r)$ represents the refractive index of the radial distance, $r = \sqrt{x^2 + y^2 + z^2}$, and R denotes the radius of the lens)¹, can be transformed into an ellipsoidal lens based on TO without affecting its properties^{19,21,22}; their refractive indices were calculated, as shown in Fig. 1. The refractive indices are distributed to compare the spherical and ellipsoidal lenses. The spherical lens has a radius $R = 50$ mm and that of the ellipsoidal lens is halved along the x-axis direction, i.e., $R_x = 25$ mm.

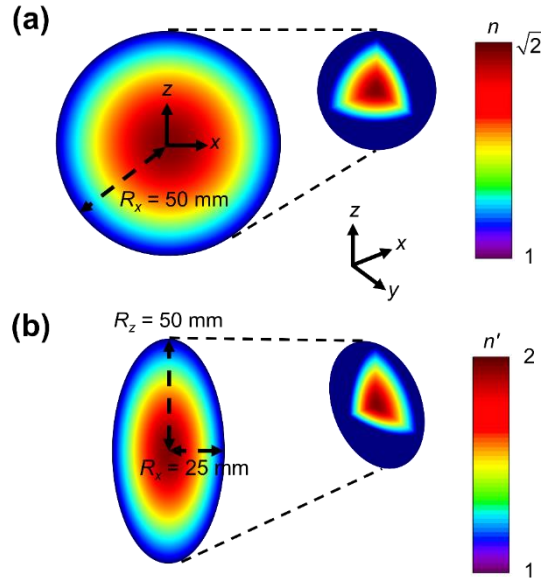


Fig. 2.1 Refractive index distributions. (a) Refractive index distribution of the spherical lens. (b) Refractive index distribution of the ellipsoidal lens. The radius of the spherical lens is 50 mm and that of the ellipsoidal lens based on TO is 25 mm in the x -axis direction. The lenses satisfy the refractive indices $1-\sqrt{2}$ and $1-2$, respectively. The color bars indicating the refractive indices are located at the top and bottom right of the figure; the insets illustrate the 3D distributions.

Further details on the TO are introduced. The transformation formula is defined to reduce and compress the radius of the x -axis direction as $x' = (\delta / R)x$, $y' = y$, and $z' = z$, where δ represents the compression factor and takes any real integer¹⁶. A spatially varying mass density and refractive index are expressed as $\rho(r) = 2 - r^2 / R^2 = n^2(r)$, and the lenses are supposed to be incompressible (bulk moduli B and B' are assumed to be approximately equal to 1 in this case). The mass density of the ellipsoidal lens can be expressed as a Jacobian matrix J combined as $\overset{=}{\rho}' = J \overset{=}{\rho} J^T / \det(J)$, where $\overset{=}{\rho}$ and $\overset{=}{\rho}'$ represent the mass density tensors of the spherical and ellipsoidal lenses; J^T and $\det(J)$ represent the transpose and determination of J , respectively^{16,21}. When the effective mass density is less than 1, the meta-atoms resonate, which results in a narrower frequency bandwidth and increased energy loss. Thus, ρ'_{yy} and ρ'_{zz} are selected as 1 without affecting their imaging performances by focusing and avoiding

the high and low refractive index regions of ρ' -map; however, there are exceptions for the some particular incident angles, such as $\pm 90^\circ$ ¹⁴⁻¹⁶. Therefore, the transformed mass density can be obtained as $\rho'_{xx} = \delta[2 - \{(\delta x')^2 + y'^2 + z'^2\} / R^2]$ ^{16,17,20-22}. Here, since $\delta = 2$, the radius of the ellipsoidal lens in the x -axis direction is halved for that of the spherical lens. The refractive indices of the spherical and ellipsoidal lenses are derived from the formulas range from $1 - \sqrt{2}$ and $1 - 2$, as shown in Fig. 1a,b, respectively. This method and the assumptions lead to the realisation of broadband metalenses as they have nondispersive meta-atoms. Further, this makes the thickness of the ellipsoidal metalens thinner than that of the spherical metalens, resulting in more efficient metalenses and providing insights for acoustic devices in applications such as medical ultrasonography, nondestructive testing (NDT), and sound navigation and ranging (SONAR).

2.2 Dispersion relation and designed metalenses

It is necessary to calculate the effective acoustic parameters of meta-atoms to be placed at each layer because the metalenses are discretised to satisfy the distributions of the refractive indices (or mass densities). The PC band structure calculations³⁹⁻⁴¹ (Supplementary Fig. 1) and S-parameter retrieval method⁴²⁻⁴⁶ (Supplementary Fig. 2) were introduced to analyse and predict the imaging performance of 3D GRIN metalenses. A multiple scattering theory (MST) method with the effective refractive index formula where Pf denotes the packing factor and subscripts α and β represent the scatterers and background, respectively; the MST method requires loosely packed inclusions, and therefore, it is excluded because it cannot meet the index of the metalens^{47,48}. The material of the metalenses is chosen stainless steel ($\rho = 8000 \text{ kg/m}^3$, $E = 200 \text{ GPa}$, and $\nu = 0.27$) for the numerical simulations and the experiments. The meta-atoms are combined with cylindrical supporters in the x -, y -, and z -axes directions (radius = 0.5 mm). Similarly, the ellipsoidal meta-atoms are joined by the cylinder in the x -axis direction and the elliptical supporters in the y - and z -axes directions. The 3D structures can be

directly fabricated with a selective laser sintering 3D printer because of the supporters. Further, the complicated structure and the structural stability increase the refractive index and impedance³⁰.

In Fig. 2, the dispersion curves, effective refractive indices, and EFCs (or EFSs) were calculated using the PC band structure calculations (Supplementary Fig. 1). The lattice constant a of the simple cubic lattice (CUB) comprising the spherical metalens is 5 mm. The length in the x-axis direction of the tetragonal lattice (TET) is 2.5 mm, which is half of the lattice constant of the CUB. Dispersion curves of the 1st acoustic band (Γ -X direction) of the irreducible Brillouin zone (IBZ) were obtained as shown in the insets of Figs. 2a and 2d. The meta-atoms were calculated by sweeping the packing factor P_f . The black dashed lines represent the energy bands in water where there is no dispersion during wave propagation. The dispersion relation in a homogeneous medium is expressed as $\omega(k) = ck$ where $\omega(k)$ represents the angular frequency dependent on the wavenumber k , c represents the speed of sound in water, and n_{eff} represents the effective refractive index. Based on this relation, n_{eff} is calculated, as shown in Figs. 2b and 2e. The higher the packing factor, the narrower the frequency band is, and the higher the effective refractive indices. The meta-atoms placed in each layer were selected such that the metalenses satisfy the effective refractive indices. However, the ellipsoidal meta-atoms do not satisfy the refractive index of 2; therefore, it is designed to satisfy the maximum effective refractive index of 1.9.

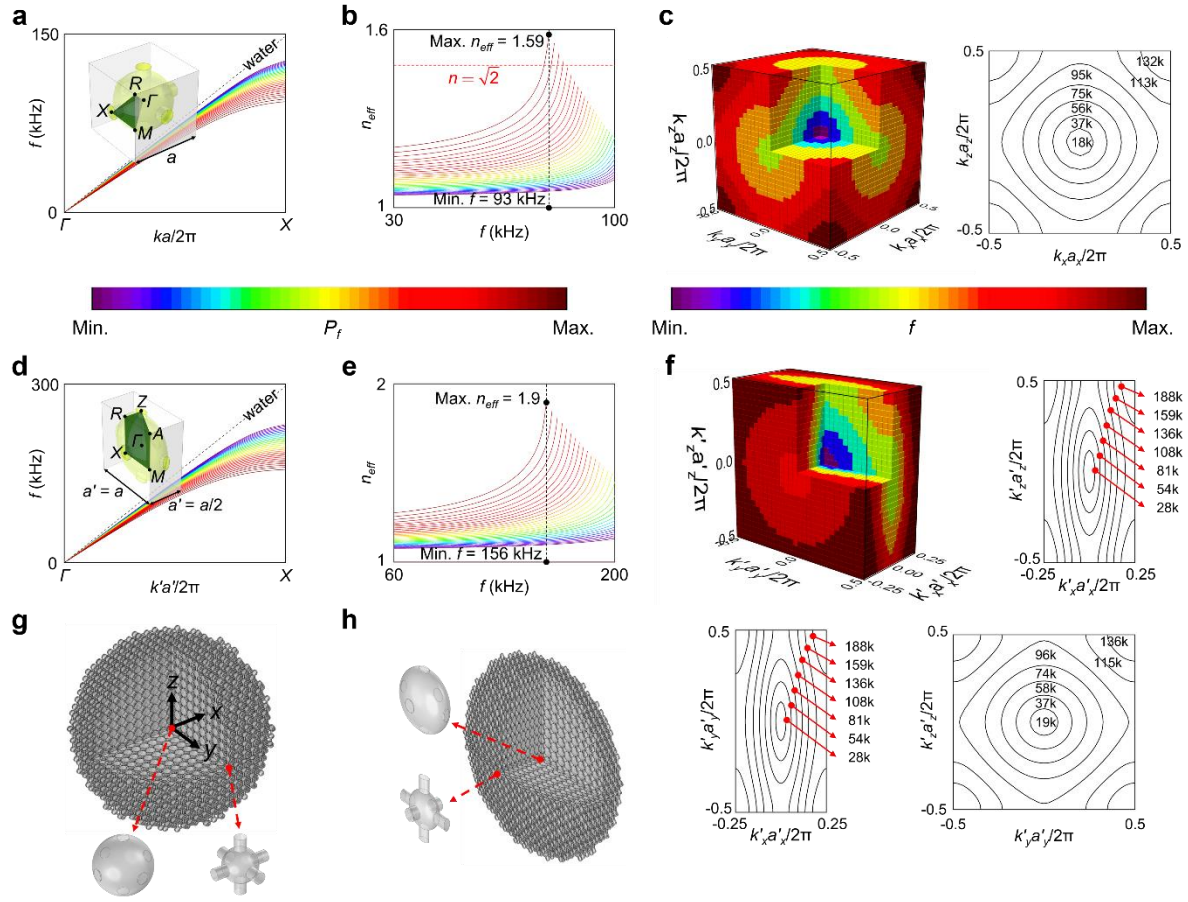


Figure 2.2 Dispersion curves, effective refractive indices, and EFCs (or EFSs) using the PC band structure calculations, and designed 3D GRIN metalenses. a,d, Dispersion relations of the spherical (a) and ellipsoidal meta-atoms (d) according to packing factor P_f . Dispersion curves are calculated along the 1st acoustic band (Γ -X direction). Black dashed lines represent the dispersion curves of energy bands in the water. The insets show PC band structures with meta-atoms (yellow) immersed in water (white); green parts represent the IBZ of the CUB (a) and TET (d). The lattice constants of the CUB and TET are 5 mm and 2.5 mm, respectively. b,e, Effective refractive indices of the spherical (b) and ellipsoidal meta-atoms (e) derived by the dispersion relation. The black circles represent the maximum refractive indices and the minimum frequencies. The red dashed line represents the refractive index of the Luneburg lens. c,f, EFSs and EFCs of the bulkiest spherical (c) and ellipsoidal meta-atoms (f) at the centre of the metalenses. All values are calculated using the PC band structure calculations. g,h, The designed spherical and ellipsoidal metalenses, and their meta-atoms (largest and smallest). These metalenses have meta-atoms that are varied mass densities of the spherical and ellipsoidal meta-atoms at each unit cell by radial distance, and combined with supporters in the x-, y-, and z-axes directions.

The frequency range is considered until the dispersion curve is nearly linear because the spherical metalens is homogeneous and isotropic. It is necessary to closely examine the focus characteristics

because the ellipsoidal metalens is not homogeneous and isotropic. The EFCs and EFSs are analysed to determine the behaviour of the wave governed by the dispersion relations, as shown in Fig. 2c,f. The EFCs are curves connecting the wavevectors of the propagating modes, and they demonstrate the focusing (or imaging) because the waves are determined by the shape of the curves. Therefore, the imaging performance is related to the slope and curvature of the EFCs in the wavevector k space for transmitting waves along a specific direction. Further, they are calculated using the PC band structure calculations, in which case the entire structures (CUB and TET) are scanned. Group velocity which is perpendicular to the EFCs in a lossless medium and specifies the direction of energy flow for the plane wave, is introduced to analyse the EFCs. The wave diffraction and propagation patterns based on the shapes of the EFCs can be divided into zero, normal (positive), and anti (negative) diffractions⁴⁹. Each shape is parallel, convex, and concave in the direction of the wave propagation. The EFCs of the spherical meta-atom are homogeneous up to a frequency of 95 kHz in Fig. 2c, whereas those of the ellipsoidal meta-atom are not homogeneous from 108–188 kHz in the k_x vs. k_z and k_x vs. k_y space, as shown in Fig. 2f. However, the EFCs of the ellipsoidal meta-atom exhibit positive diffractions up to 159 kHz and anti-diffractions from 188 kHz. The minimum frequencies for the spherical and ellipsoidal meta-atoms with maximum refractive indices of 1.59 and 1.9, respectively, are 93 kHz and 156 kHz. The upper and lower limits of available frequencies for the spherical and ellipsoidal metalenses are approximately 100 kHz and 160 kHz, and 30 kHz and 60 kHz, respectively, because the wavelengths of these frequencies are smaller than the radius of the metalenses for interacting with the acoustic waves. The designed spherical and ellipsoidal metalenses are discretised and divided into 10 concentric layers, as shown in Fig. 2g,h. The refractive indices were obtained to be increased from the surface to the centre.

2.3 Retrieved values

As a part of further steps to verify how the metalenses work, the S-parameter retrieval method, which is extended from the effective parameters of electromagnetic materials³⁰⁻⁴³ to acoustic metamaterials⁴⁴, was performed by normally incident plane waves on the PC band structure, and by calculating S_{11} and S_{21} , which are reflection (R) and transmission (T) coefficients, respectively

(Supplementary Fig. 2). The effective refractive index and acoustic impedance are expressed as

$$n_{\text{eff}} = [\pm \cos^{-1}(\{1 - (S_{11}^2 - S_{21}^2)\} / 2S_{21}) + 2\pi m] / ka \quad \text{and} \quad Z_{\text{eff}} = \pm \sqrt{\{(1 + S_{11})^2 - S_{21}^2\} / \{(1 - S_{11})^2 - S_{21}^2\}},$$

where m represents the branch number of $\cos^{-1}$, k denotes the wavenumber, a represents the slab thickness (or lattice constant), and “ $\pm$ ” restricts the imaginary part of the index $\text{Im}(n_{\text{eff}})$ to be negative⁴⁴.

When $\text{Re}(Z_{\text{eff}})$ or $\text{Im}(n_{\text{eff}})$ is close to zero, a computational error can cause incorrect combinations of the signs. Thus, it is redefined as $n_{\text{eff}} = (-i \log x + 2\pi m) / kd$ and $Z_{\text{eff}} = r / (1 - 2S_{11} + S_{11}^2 - S_{21}^2)$, where

and $x = (1 - S_{11}^2 + S_{21}^2 + r) / 2S_{21}$ and $r = \mp \sqrt{(S_{11}^2 - S_{21}^2 - 1)^2 - 4S_{21}^2}$, the effective bulk moduli B_{eff}

and the effective mass densities ρ_{eff} are defined as $B_{\text{eff}} = Z_{\text{eff}} / n_{\text{eff}}$ and $\rho_{\text{eff}} = Z_{\text{eff}} \times n_{\text{eff}}$. The effective

acoustic material properties (speed of sound, acoustic impedance, bulk modulus, and mass density) are shown in Fig. 3.

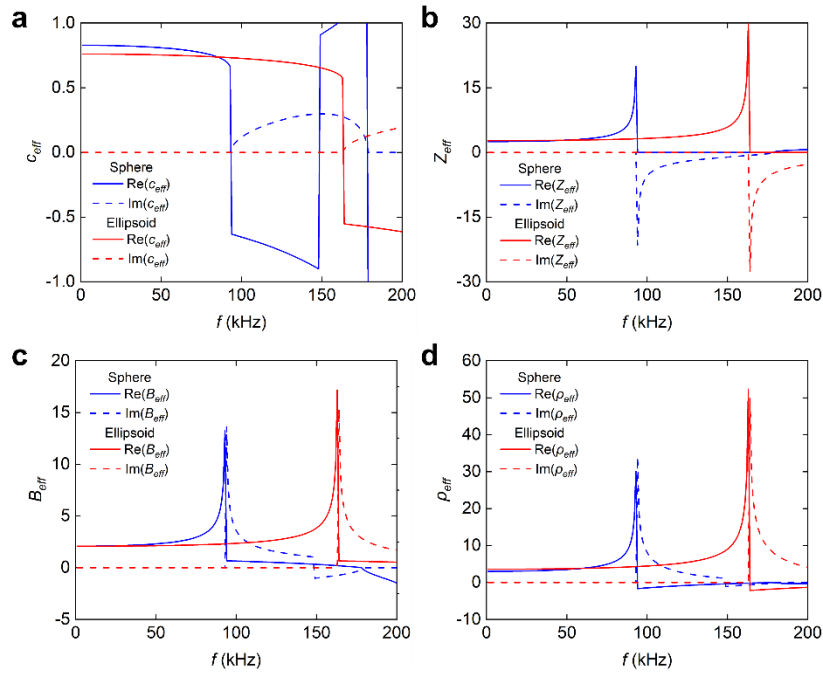


Figure 2.3 a-d, Real (solid line) and imaginary part (dashed line) of the effective speed of sound c_{eff} , acoustic impedance Z_{eff} , bulk modulus B_{eff} , and mass density ρ_{eff} of the spherical (blue) and ellipsoidal meta-atoms (red) are plotted in the frequency range of 1–200 kHz. All values are calculated using the S-parameter retrieval method (Supplementary Fig. 2a,b).

The largest meta-atoms were selected for the retrieved values because the larger the volume of meta-atoms, the lower is the phononic band gap (PBG) frequency, as indicated in Fig. 2 (PBG of the bulkiest spherical meta-atom is shown in Supplementary Fig. 1b). In Fig. 3a, the real parts of the effective speed of sound c_{eff} of the spherical and ellipsoidal meta-atoms change from positive to negative approximately 100 kHz and 160 kHz, respectively. These are meant that ultrasound undergoes negative refraction at these frequencies. In Supplementary Fig. 2c,d, the higher the frequency, the greater is the reflection value (S_{11}) than the transmission value (S_{21}). Z_{eff} , B_{eff} , and ρ_{eff} (in Fig. 3b–d) are increased when the frequency is increased. The spherical and ellipsoidal meta-atoms have negative effective parameters because the acoustic waves are completely blocked near 100 kHz and 160 kHz, respectively. These parameters are designed with nonresonant scatterers, and the effective parameters calculated by the S-parameter retrieval method agreed well with the results analysed by the PC band structure calculations (Supplementary Fig. 1c,2b). For the designed acoustic metamaterials, the regions corresponding to PBG, and negative properties are excluded from the available frequencies.

2.4 Underwater ultrasound imaging

The imaging and beam steering performances of the 3D GRIN metalenses are shown in Fig. 4–5. The underwater acoustic metalenses were simulated (Supplementary Fig. 3), fabricated (Supplementary Fig. 4), and experimented on (Supplementary Fig. 5) in water ($\rho = 998 \text{ kg/m}^3$ and $c = 1,481 \text{ m/s}$). The scanned area is shown in Fig. 4a. They were analysed for the focal length and the FWHM in the scanned area. The focal length is the distance from the surface of the metalenses to the point where the highest sound intensity is obtained along the centre line of the metalenses. The FWHM is defined as the width of the spectral sound intensity curve obtained between the points on the y-axis of the sound intensity amplitude, which is half of the maximum sound intensity, $I_{max}/2$. All values of the sound intensity are normalised from 0 to 1, and the white region indicates the area where the ultrasound is focused. When the plane waves are normally incident on the metalenses, the xy- ($60 \times 120 \text{ mm}^2$) and yz- ($120 \times 120 \text{ mm}^2$) planes are scanned to obtain the sound pressure and their images. The incident angles from 0° to

centre. However, the spherical metalens shows uniform performance in all directions (omnidirectionality). Therefore, Figure 4b and 4c (especially in the xy -plane) show that the spherical metalens has a relatively better beam steering performance than that of the ellipsoidal metalens. The graph demonstrating these results in more detail is shown in Fig. 4d. The ellipsoidal metalens has a longer focal length than that of the spherical metalens (approximately $0\text{--}0.5\lambda$) and is inconsistent. The FWHM of the ellipsoidal metalens is slightly over subwavelength imaging at the incident angles of 40° and 10° in the numerical and experimental results, respectively. The FWHM of the spherical metalens is almost flat near 0.6λ .

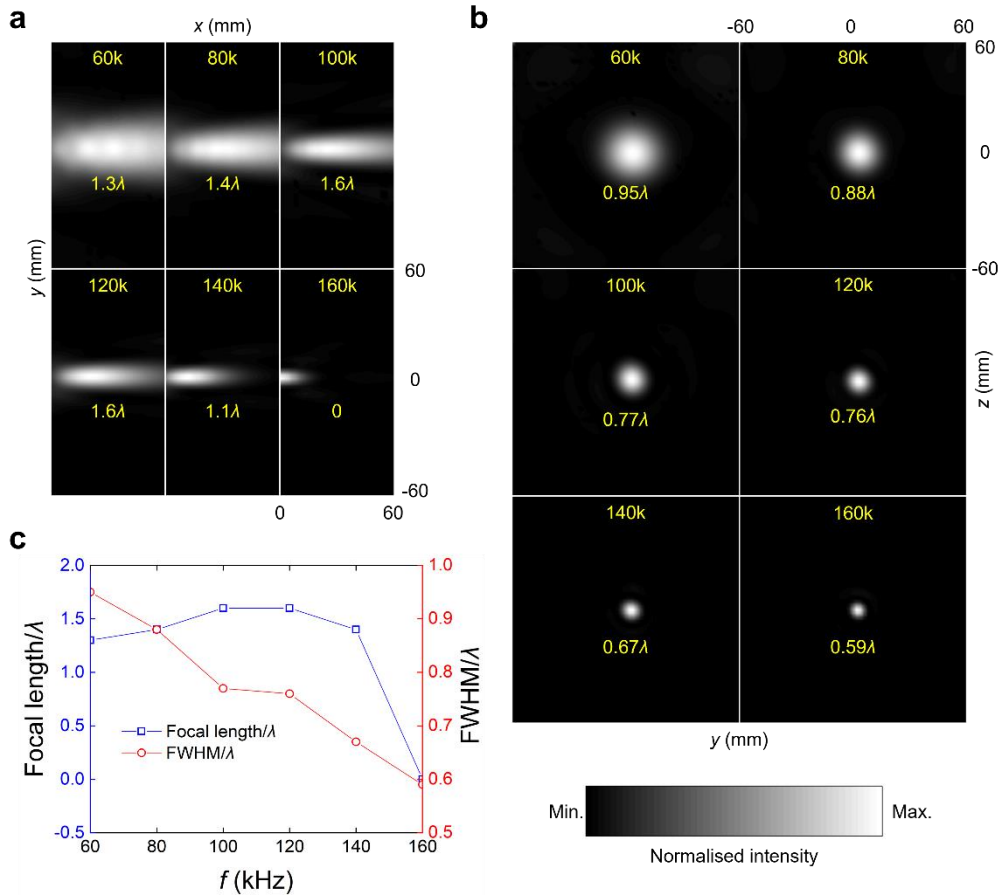


Figure 2.5 a,b, Experimental results of the underwater ultrasound subwavelength imaging of the ellipsoidal metalens in the xy - and yz -planes, respectively. Measurements were performed at frequencies between 60 kHz and 160 kHz in 20 kHz steps when the incident angle θ is zero. The numerical and experimental methods and expressions are the same in Fig. 4a. c, Comparison of the experimental results for the focal length and FWHM by frequency. The blue line-hollow square

and red line-hollow circle represent the focal length and FWHM, respectively.

In Fig. 5, a more detailed image of the ellipsoidal metalens is analysed experimentally. Figure 5a and 5b show imaging in the xy- and yz-planes at frequencies between 60 and 160 kHz in 20 kHz steps, respectively, when the incident angle θ is zero. As indicated by the previously analysed properties of meta-atoms, subwavelength imaging was achieved at an available frequency of 60 kHz to 160 kHz. A frequency-dependent imaging performance was confirmed intuitively. The values of the focal length and FWHM tended to decrease with increasing frequency, as indicated Fig. 5c. Consequently, TO-based metalens has a wide bandwidth at high frequencies for underwater ultrasonic waves.

Chapter 3. SUPPLEMENTARY INFORMATION

3.1 Numerical simulations

The finite element method software, COMSOL Multiphysics, was used to simulate the coupled acoustic pressure and structural mechanics, as shown in Supplementary Fig. 1–3. The PC band structure calculations were performed by eigenfrequency analysis, and the Floquet-Bloch boundary condition was applied to the PC band structure (Supplementary Fig. 1a). Thus, PBG, Bloch modes (1 and 2), and EFC (and its projection) were demonstrated, as shown in Supplementary Fig. 1b–d. The PBG, which blocks wave propagation, was formed between in the frequency range of 93–187 kHz, and A (blue point) and B (red point) are indicated at the ends of the dispersion curve (X). Bloch mode 2 corresponding to PBG was obtained in B. EFC supports these results because of the homogeneous frequency range, as shown in Supplementary Fig. 1d. An analysis was performed to calculate the effective acoustic parameters using the S-parameter retrieval method in the frequency domain in Supplementary Fig. 2a. The results are plotted in Supplementary Fig. 2b–d. Modes such as Bloch mode 1 (Supplementary Fig. 1c) were calculated at a frequency of 80 kHz in Supplementary Fig. 2b. In this case, the reflection of the acoustic wave is greater than the transmission (Supplementary Fig. 2c,d), which in turn is useful for verifying the frequency range for positive acoustic metamaterials. Plane waves propagate from both ends of the boundary to numerically demonstrate the imaging performance of the metalenses numerically, as indicated in Supplementary Fig. 3a. The boundaries of all meta-atom supporters in

the x -, y -, and z -axes directions were completely fixed. The focal length and the FWHM were achieved, as shown in Supplementary Fig. 3c,d, respectively, through the underwater ultrasound imaging shown in Supplementary Fig. 3b; they were scanned along the y - and x -axes directions, respectively. The numerical and experimental results were in good agreement. Here, it is introduced representatively for the spherical metalens and its meta-atom.

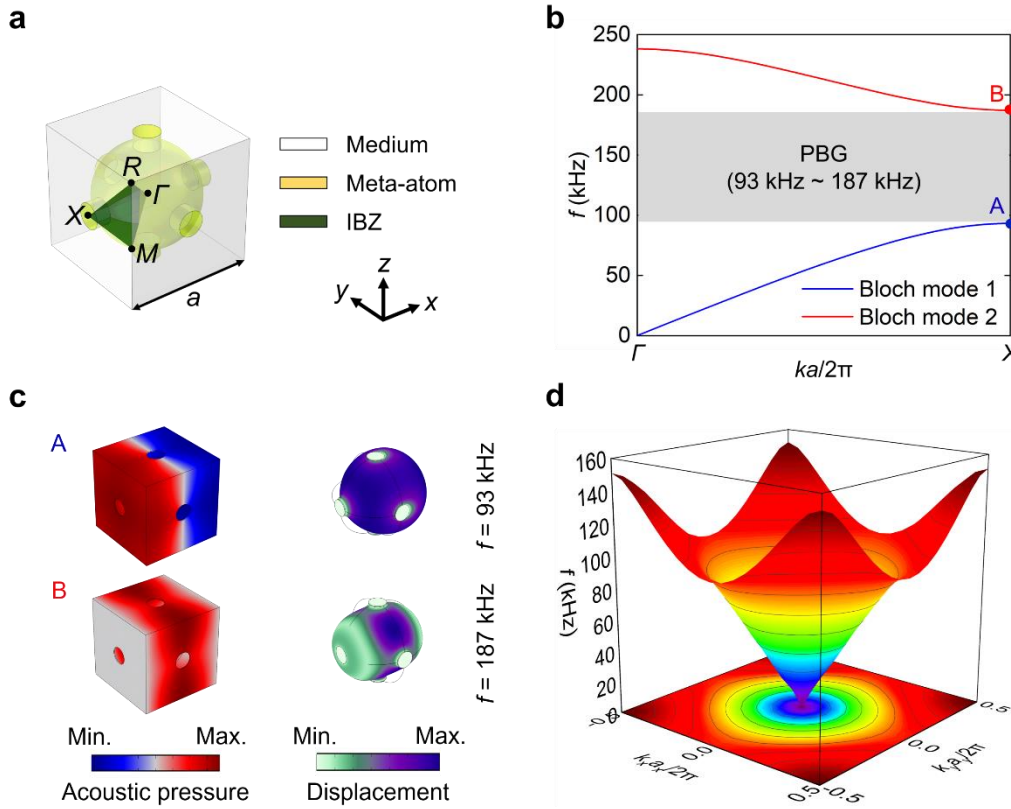


Figure 2.6 a, PC band structure of the sphere meta-atom combined with the cylindrical supporters in the x -, y -, and z -axes directions. White, yellow, and green represent medium (water), meta-atom (sphere), and IBZ, respectively. b, PBG of the first band (Γ -X) dispersion curve. c, Numerical results for the acoustic pressure and displacement of the PC band structure. d, EFC and its projection. All simulations are performed on the bulkiest meta-atom. PBG (b) is formed between the Bloch modes 1 and 2 at frequencies between 93 kHz (point A) and 187 kHz (point B), and then the numerical results (c) are introduced the Bloch modes at points A and B for the acoustic pressure in the medium and displacement of meta-atom, respectively. EFC (d) is shown when k_z is 0, and k_x and k_y are normalised.

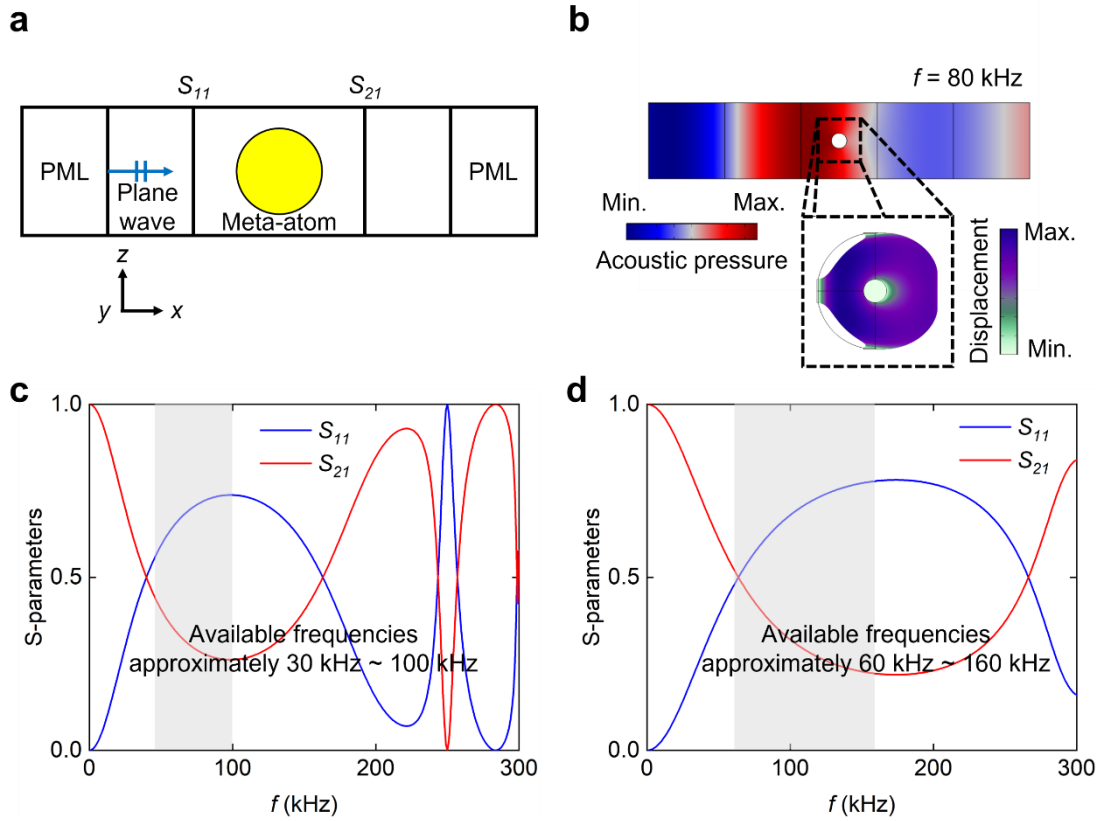


Figure 2.7 a,b, Schematic diagram (a) and numerical simulation results (b) using the S-parameter retrieval method introduced to calculate the effective material properties of meta-atoms. Acoustic plane waves are normally incident on the meta-atoms from left to right, and perfectly matched layers (PML), which cancel out reflection waves, are set at both ends. The S-parameters are calculated at the boundary of the PC band structure. In this case, the bulkiest spherical meta-atom is shown as a representative at a frequency of 80 kHz. **c,d,** The S-parameters S_{11} and S_{21} refer to reflection (R) and transmission coefficients (T), respectively. These graphs show the numerical simulation results for the spherical (c) and ellipsoidal meta-atoms (d) at frequencies from 1 kHz to 300 kHz. The retrieved values are normalised.

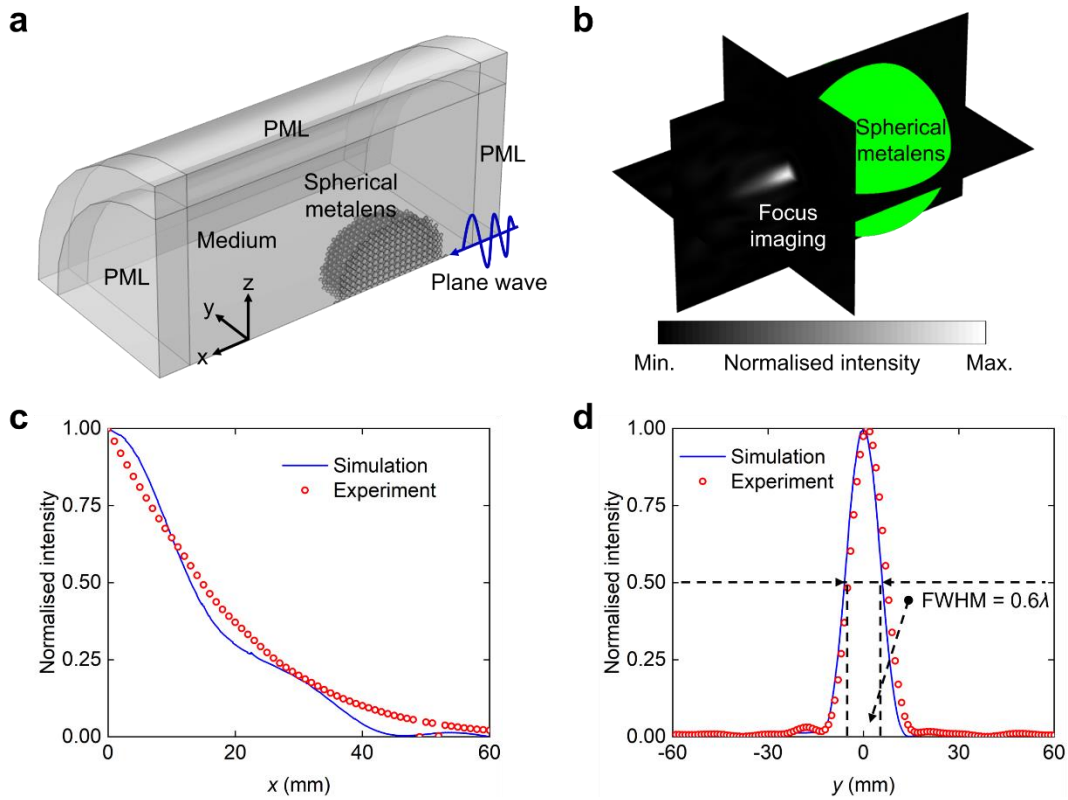


Figure 2.8 a,b, Simulation modeling (a) and numerical results (b) of the spherical metalens in water. **c,d**, Normalised intensity curve of numerical (solid blue lines) and experimental results (red hollow circles) of the spherical metalens on the y- (c) and x-axis (d) to calculate the focal length (the distance from the surface of the metalenses to the point where the highest sound intensity is obtained along the centre line of the metalenses) and FWHM (the width of the spectral sound intensity curve obtained between the points on the y-axis of the sound intensity amplitude, which is half of the maximum sound intensity, $I_{max}/2$) when the incident angle is 0° , and the frequency is 80 kHz. The sound intensity values are normalised.

3.2 Fabrication

The 3D GRIN metalenses made of stainless steel were fabricated using a selective laser sintering 3D printer. The fabricated spherical and ellipsoidal metalenses are shown in Supplementary Fig. 4a,b, respectively. Their lens supporters (area circled in red) are coupled at the bottom to fix the metalenses to the testbed frame. Since the fabrication resolution is 0.1 mm, the fabricated metalenses may not satisfy the performance of the designed metalenses.

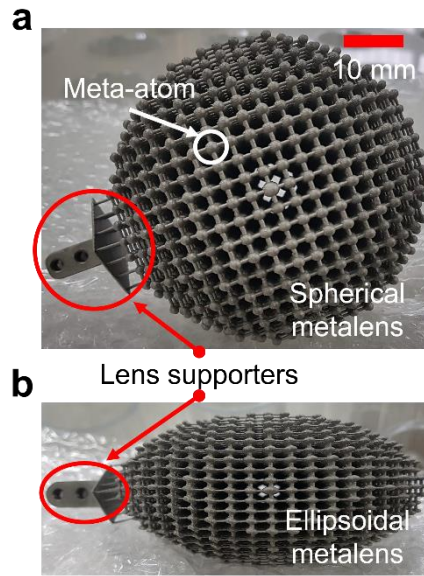


Figure 2.9 a,b, Fabricated 3D GRIN metalenses made of stainless steel using selective laser sintering 3D printer. They are the spherical (a) and ellipsoidal metalenses (b) combined with the lens supporters that can be fixed to the testbed.

3.3 Experimental setup, method, and measured signals

The burst signal graph before and after passing through the spherical metalens is shown in Supplementary Fig. 5d. The upper black line (input burst signal) represents the waveform at the point entering the metalens, and the other line (output burst signal) represents the waveform at that, which is the maximum amplitude passing through the metalens. A function generator was built into the measurement program, and the parameters were set as follows: burst period = 50 ms, sensitivity = 3.16 mV/Pa, wait-time = 300 ms, move-time = 300 ms, and signal wait-time = 100 ms. We set the position to 200 μ s and the base to 100 μ s/div and processed 8192 samples at a rate of 7.6923 MHz. The burst signal was applied to the metalenses to minimise the reflection of sound waves by the metalenses or the surrounding objects. However, the output burst signal noise was measured owing to such effects. The metalenses increase the amplitude of the output burst signal by approximately 2 times compared to that of the input burst signal, and it can be seen intuitively that the metalenses amplify the waves.

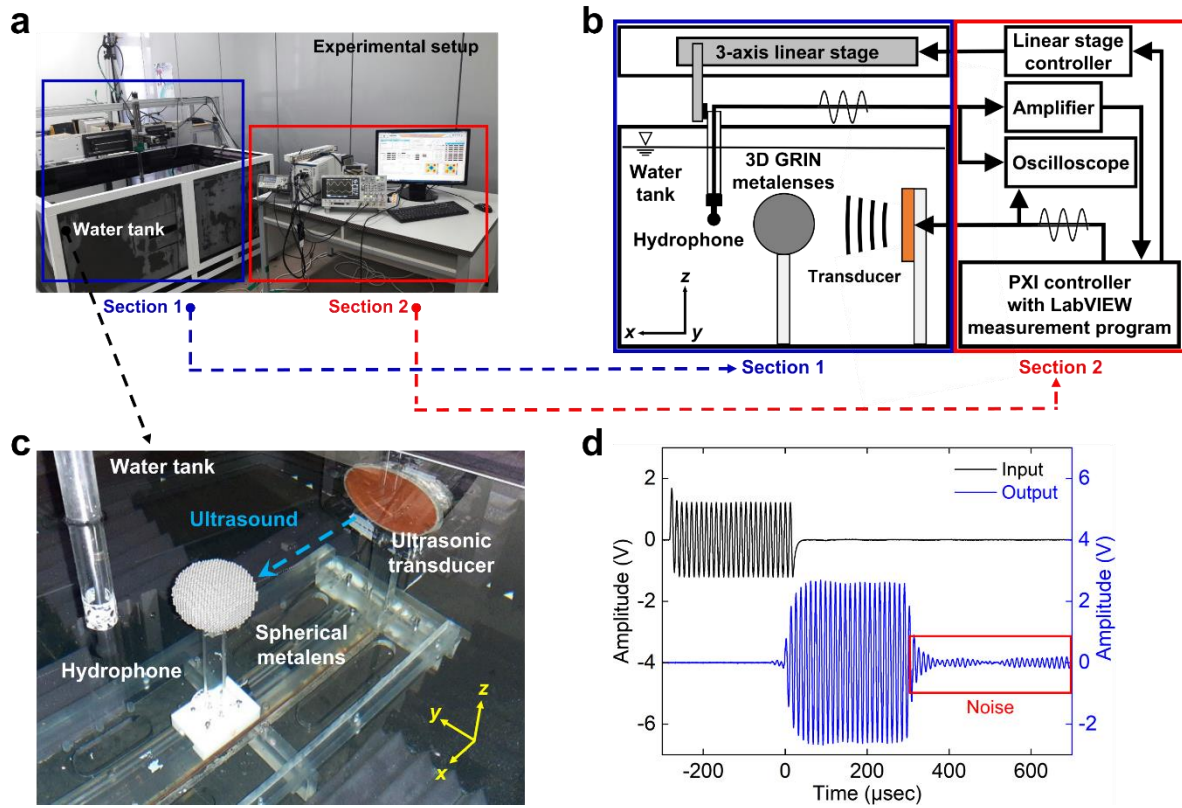


Figure 2.10 a-c, Experimental environment for measuring the ultrasonic waves in water. Experimental setup (a) and its schematic diagram (b). Several pieces of equipment such as a three-axis linear stage and its controller, PXI controller with LabVIEW measurement program, oscilloscope, hydrophone, transducer, and amplifier are used. Setting for measuring underwater ultrasound using the hydrophone (receiving), 3D GRIN metalenses (spherical metalens in the figure), and ultrasonic transducer (transmitting) in the water tank (c). **d**, Burst signals before and after passing through the spherical metalens at a frequency of 80 kHz. The red solid line box marks the area where the noise is measured.

Chapter 4. DISCUSSION

In this study, the numerical simulations and experiments of 3D GRIN metalenses based on the Luneburg lens and TO were performed underwater. The shape of the underwater acoustic metalenses was divided into the spherical and ellipsoidal types, and their imaging performances were verified. The dispersion curves, EFCs (or EFSs), and effective material properties were calculated using the PC band structure calculations and S-parameter retrieval method, and the Bloch modes of meta-atoms were in

good agreement. The wavevectors underwent positive refraction through their EFCs, and frequency ranges were identified in more detail. Further, ultrasound subwavelength focus imaging can be achieved numerically and experimentally in water.

However, some future works are required: (1) Stainless steel is used to achieve a high impedance difference of the metalenses to water. However, air inclusions may be an alternative. Air enters the inside of scatterers, which is ideal as it can generate a high impedance difference even if the structure is not metallic^{31,32}. It is also an economical choice in terms of cost. If TO is applied to the metalenses composed of these meta-atoms, they can be designed and work efficiently. (2) Commercializing these metalenses immediately is not easy to fabricate because of their complex structure. This is a limitation of GRIN metalenses (and probably all metamaterials). Thus, it is necessary to design a structure that is as simple as possible and can effectively maintain performance. (3) The beam steering performance of the ellipsoidal metalens is deteriorated, as shown by the EFCs (or EFSs) in Fig. 2f and the imaging in Fig. 4. Further, the imaging performance may be limited in the case of a high incident angle such as 90° . This is inevitable because it deviates from the spherical shape, which has the same performance for all incident angles.

Despite these shortcomings, the advantages of each metalens are clear and should be emphasised. The spherical metalens is relatively bulky than the ellipsoidal metalens but is omnidirectional and does not suffer from aberration. The ellipsoidal metalens will be suitable for broadband devices because the available frequency range is broader than that of the spherical metalens. If various structures suitable for use are proposed by applying TO to metalenses, these metalenses are expected to have prospects in imaging applications requiring a wide FOV such as medical imaging systems, NDT, and SONAR in the future.

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